

EXPT NO:4	TRANSFORM DATA TO SUIT BI REPORTING REQUIREMENTS AND DESIGN A DASHBOARD WITH KEY PERFORMANCE INDICATORS (KPIs) USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the role of data transformation in BI reporting?

Data transformation converts raw data into a clean, organized, and structured format for analysis. It improves data quality by correcting errors, handling missing values, changing data types, and creating calculated fields. This helps generate accurate reports and dashboards.

2. What are the common data cleaning techniques in Tableau and Power BI?

Common data cleaning techniques include:

- Removing duplicate records.
- Handling missing (null) values.
- Changing data types (e.g., text to date or number).
- Renaming columns for better understanding.
- Filtering unwanted records.
- Creating calculated fields.
- Splitting or merging columns when required.

3. How do Tableau and Power BI handle data aggregation?

Tableau and Power BI automatically aggregate numerical data using functions such as:

- SUM
- AVERAGE (AVG)
- COUNT
- MIN
- MAX

Users can also change the aggregation method according to their analysis needs.

4. What are KPIs, and why are they important in a dashboard?

KPIs (Key Performance Indicators) are measurable values used to evaluate performance or progress toward a goal. They provide quick insights, help monitor important metrics, support decision-making, and make dashboards easier to understand.

5. What are the different types of charts available in Tableau and Power BI for visualization?

Both Tableau and Power BI provide various visualization types, including:

- Bar Chart
- Column Chart
- Line Chart
- Area Chart
- Pie Chart
- Donut Chart
- Scatter Plot
- Map
- Tree Map
- Heat Map
- Table
- KPI Card
- Gauge Chart
- Funnel Chart
- Bubble Chart
- Waterfall Chart

These charts help visualize trends, comparisons, distributions, relationships, and geographic data effectively.

IN-LAB EXERCISE:

OBJECTIVE:

To transform COVID-19 data into a structured format suitable for Business Intelligence (BI) reporting and design interactive dashboards with Key Performance Indicators (KPIs) using Tableau and Power BI.

PROCEDURE:

1. Data Collection:

- Import the **COVID-19 dataset** from inbuilt sources like *Our World in Data* or CSV files containing country-level COVID statistics (e.g., new cases, deaths, vaccinations).
- Datasets may include fields such as *date*, *location*, *new_cases*, *total_cases*, *new_deaths*, *total_deaths*, *people_vaccinated*, etc.

2. Data Cleaning and Transformation:

- Convert columns like *date* to proper date formats.
- Handle missing or null values in *new_cases*, *total_cases*, etc.
- Create calculated fields (e.g., daily case growth rate, case fatality rate, vaccination percentage).

3. Data Aggregation:

- Aggregate data by *continent*, *country*, or *month*.
- Use measures like SUM(*new_cases*), AVG(*positivity_rate*), MAX(*total_deaths*).

4. Designing KPIs:

- Identify KPIs such as:
 - Total Confirmed Cases
 - Total Deaths
 - Case Fatality Rate
 - Vaccination Coverage (%)
 - Daily Case Growth Rate
- Use DAX (Power BI) or Calculated Fields (Tableau) to compute and visualize these KPIs.

5. Dashboard Development:

- Build visualizations such as:
 - Line charts for daily new cases and deaths
 - Bar charts for country-wise vaccination progress

- Maps to show regional case distribution
- KPI cards for high-level insights
- Use filters (e.g., by continent, country) and slicers (e.g., by date range) for interactivity.

6. Publishing and Sharing:

- Publish the dashboard using Tableau Public/Server or Power BI Service.
- Share the dashboards with stakeholders through public or organizational portals.

Experiment Dataset:

COVID-19 Global Dataset

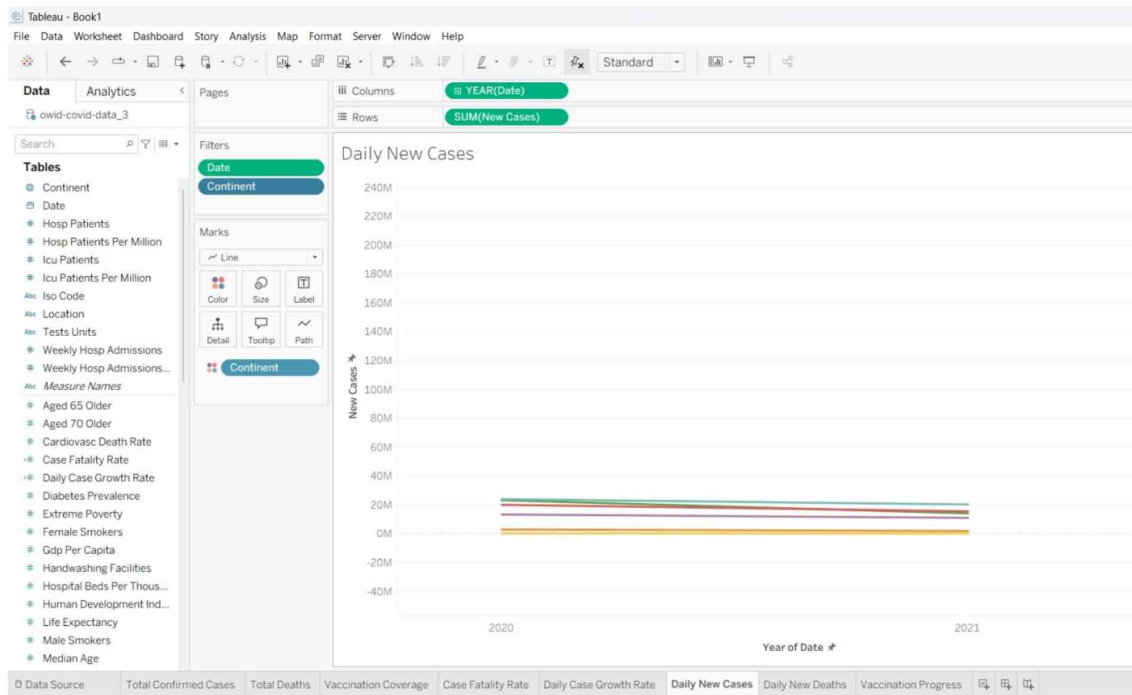
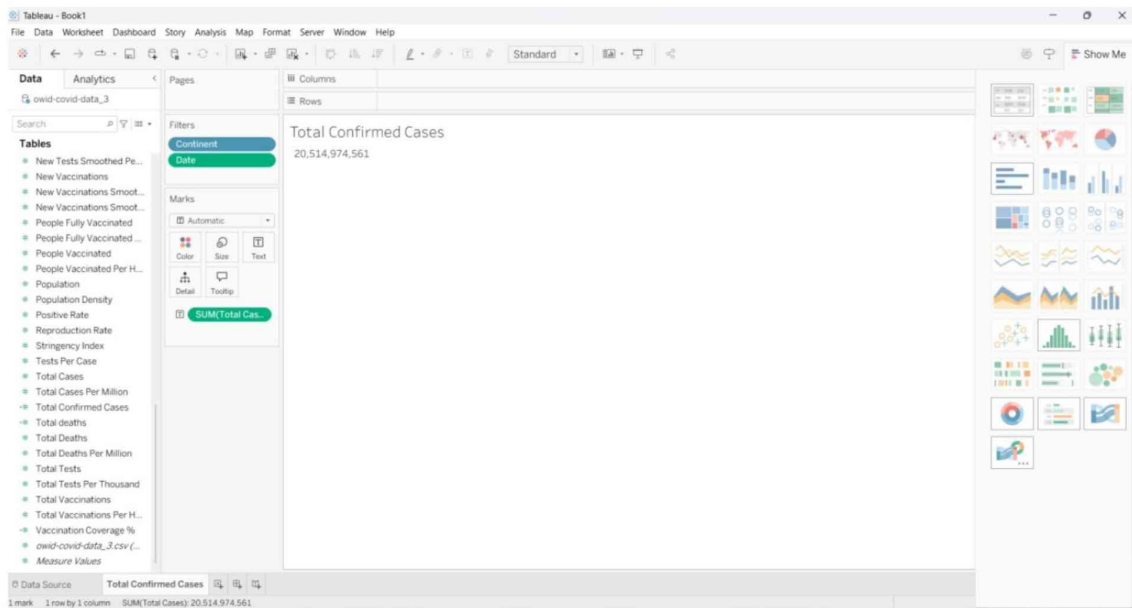
Fields Used:

- **Date** – Date of the report
- **Location** – Country or region
- **New Cases** – Daily reported cases
- **New Deaths** – Daily reported deaths
- **People Vaccinated** – Cumulative vaccinated count
- **Total Cases** – Cumulative confirmed cases
- **Total Deaths** – Cumulative deaths
- **Population** – Total population of the country

Tableau steps :

The screenshot shows the Tableau Desktop interface with the 'owid-covid-data_3' data source loaded. The 'Fields' pane on the right lists the following fields:

Type	Field Name	Physical Table	Rem...
Abc	Iso Code	owid-covid-d...	iso_co...
Abc	Continent	owid-covid-d...	contin...
Abc	Location	owid-covid-d...	location
Abc	Date	owid-covid-d...	date
*	Total Cases	owid-covid-d...	total_...
*	New Cases	owid-covid-d...	new_c...
*	New Cases Smoothed	owid-covid-d...	new_c...
*	Total Deaths	owid-covid-d...	total_...
*	New Deaths	owid-covid-d...	new_d...
*	New Deaths Smoothed	owid-covid-d...	new_d...
*	Total Cases Per Million	owid-covid-d...	total_...
*	New Cases Per Million	owid-covid-d...	new_c...
*	New Cases Smoothed I	owid-covid-d...	new_c...
*	Total Deaths Per Million	owid-covid-d...	total_...



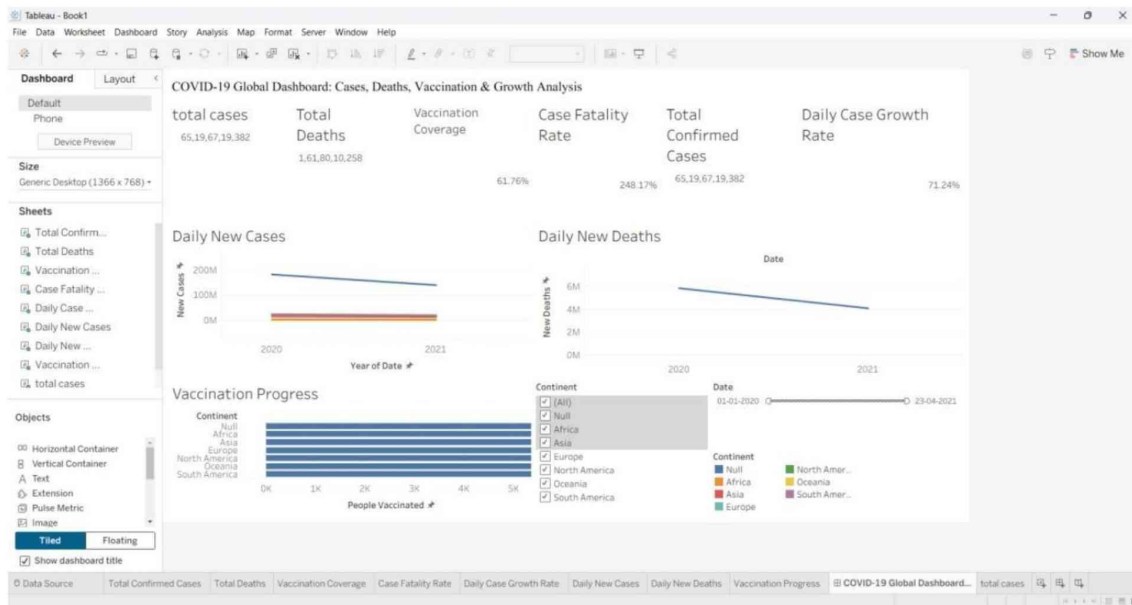
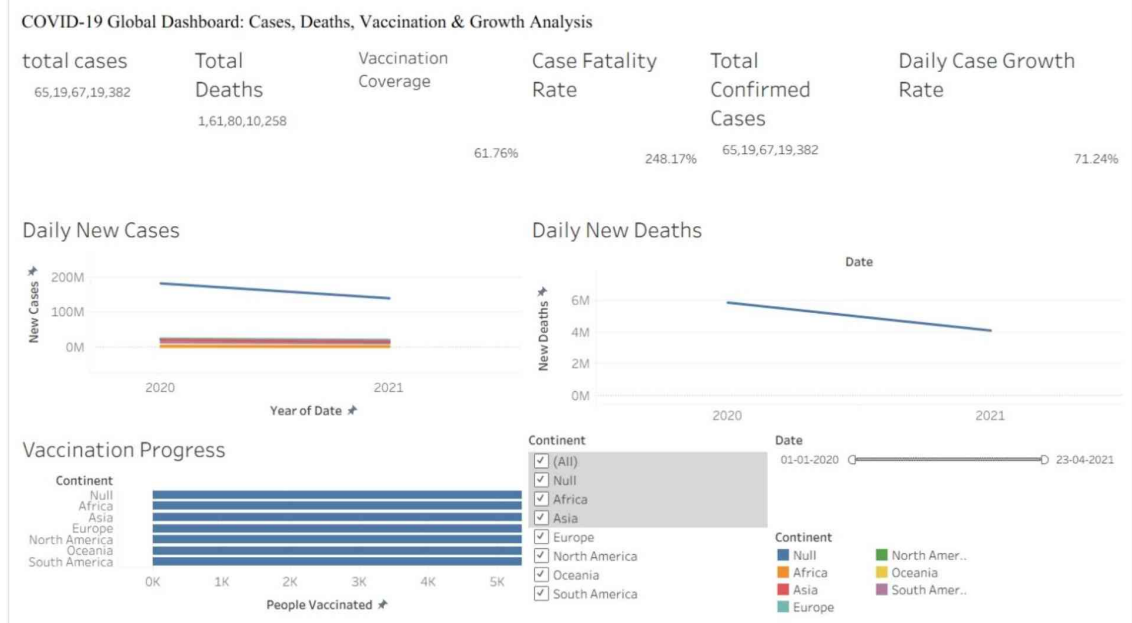


Tableau dashboard :



Power BI steps :

The first screenshot shows the initial data load into the Power Query Editor. The data is organized into columns: **id**, **iso_code**, **continent**, **location**, **date**, **total_cases**, **new_cases**, and **new_deaths**. The data represents daily COVID-19 statistics for Afghanistan from February 24, 2020, to May 26, 2020.

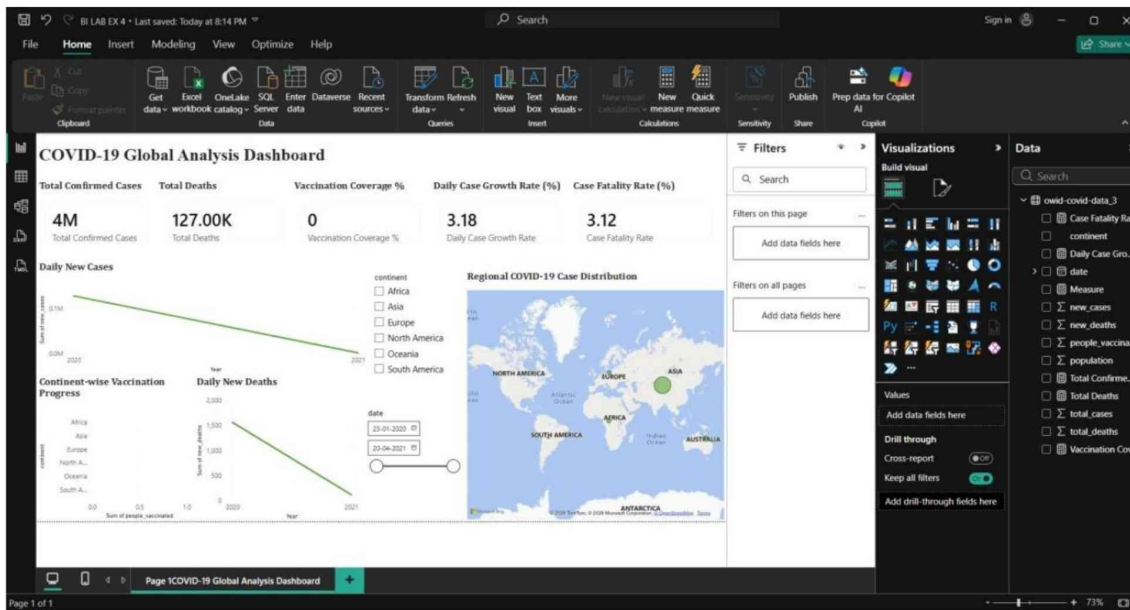
The second screenshot shows the application of a transformation step named "Removed Duplicates". The formula bar displays the M code: `Table.ReplaceValue(*Removed Duplicates*,null,0,Replacer.ReplaceValue,{"reproduction_rate"})`. The data table now includes an additional column, **reproduction_rate**, which contains numerical values representing the daily reproduction rate.

Query Settings (First Screenshot):

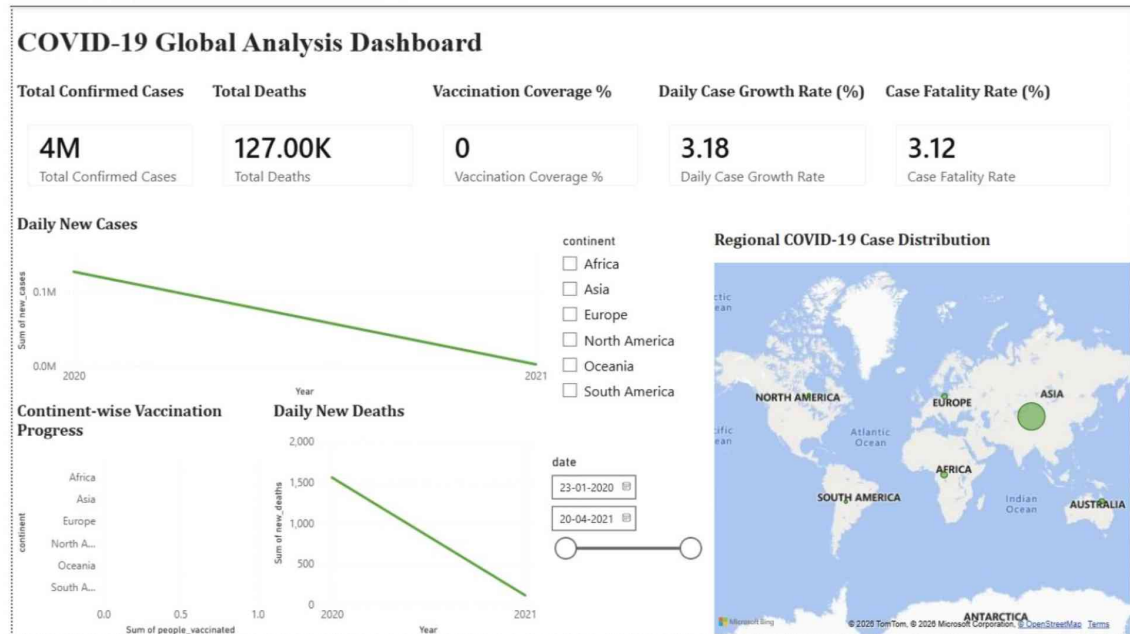
- NAME:** covid-covid-data_3
- APPLIED STEPS:** Source, Promoted Headers, Changed Type

Query Settings (Second Screenshot):

- NAME:** covid-covid-data_3
- APPLIED STEPS:** Source, Promoted Headers, Changed Type, Replaced Value, Replaced Value1, Replaced Value2, Replaced Value3, Replaced Value4, Replaced Value5, Replaced Value6, Replaced Value7, Removed Duplicates, Replaced Value8



FINAL POWER BI DASHBOARD :



Power BI dashboard link :

https://kumaragurudtsteam-my.sharepoint.com/:u:/r/personal/roshniangel_24ad_kct_ac_in/Documents/English/BI%20LAB%20EX%204.pbix?csf=1&web=1&e=CM3u60

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did data transformation improve the quality of insights in your COVID-19 dashboard?

Data transformation improved the accuracy and reliability of the dashboard by converting data into the correct format, removing inconsistencies, handling missing values, and creating calculated fields such as vaccination coverage and case fatality rate. This made the visualizations more meaningful and easier to analyze.

2. What challenges did you encounter during the data cleaning process, and how did you resolve them?

The main challenges were handling null values, correcting data types, and fixing missing values in some columns. These issues were resolved by filtering null records, converting columns to the correct data types, and using calculated fields to ensure accurate analysis and visualization.

3. Which KPI provided the most valuable insight, and why?

The **Vaccination Coverage (%)** KPI provided the most valuable insight because it showed the percentage of the population that had been vaccinated. This helped compare vaccination progress across different countries and regions and evaluate public health efforts.

4. How does interactivity in dashboards improve user experience?

Interactive dashboards allow users to filter, drill down, and explore data dynamically. Features such as filters, slicers, and maps enable users to focus on specific countries, continents, or time periods, making data analysis faster and more effective.

5. What steps would you take to further enhance the dashboard's effectiveness?

To further improve the dashboard, I would:

- Add more interactive filters and slicers.
- Include trend forecasting and predictive analytics.
- Use additional KPIs for better analysis.
- Improve the dashboard layout with clear titles and color coding.
- Schedule automatic data refresh for real-time reporting.

RESULT:

The COVID-19 dataset was successfully cleaned, aggregated, and transformed for visual analysis using Tableau and Power BI. Missing values, data type mismatches, and inconsistencies were addressed. KPIs such as vaccination rate and daily case growth offered valuable insights into the pandemic trends across countries and timeframes. The resulting interactive dashboards allowed for effective public health monitoring and communication.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		